

Academic curriculum vitae

Name: Tom Van Cutsem

Nationality: Belgian

Residence: Leuven, Belgium

Languages: Dutch (Mother Tongue), English (Fluent, CEFR level C1 accredited)



Higher Education

- 2004–2008: Doctor of Philosophy (**PhD**) in **Computer Science**, Vrije Universiteit Brussel. Dissertation: *Ambient references: object designation in mobile ad hoc networks*. Promotors: Prof. Dr. Wolfgang De Meuter, Prof. Dr. Theo D'Hondt. Grade: “met grootste onderscheiding” (lit.: *highest distinction*)
- 2000–2004: Master of Science (**MSc**) in **Computer Science** (Dutch: Licentiaat in de Informatica), Vrije Universiteit Brussel. Graduation thesis: *A prototype-based approach to distributed applications*. Promotor: Prof. Dr. Theo D'Hondt. Grade: “met grootste onderscheiding” (lit.: *highest distinction*)
- 2008: extracurricular course on Entrepreneurship, Vrije Universiteit Brussel.

Positions Held

- 2022–present: **Associate Professor** (Hoofddocent) at DistriNet KU Leuven, Leuven, Belgium.
- 2022–present: Part-time **Research Advisor** at Nokia Bell Labs, Antwerp, Belgium.
- 2016–2022: **Research Director** (Department Head) at Nokia Bell Labs, Antwerp, Belgium.
- 2014–2016: **Senior Researcher** at Nokia Bell Labs, Antwerp, Belgium.
- 2012–2014: **Assistant Professor** (Docent) at Vrije Universiteit Brussel (VUB), Brussels, Belgium.
- 2008–2014: **FWO Postdoctoral Fellow** of the Research Foundation, Flanders (personal research grant).
- 2010–2012: Part-time (10%) Assistant Professor (Docent) at Vrije Universiteit Brussel.
- 2009–2010: **Visiting Faculty** at Google Research, Mountain View, California, USA.
- 2004–2008: **FWO PhD Fellow** at VUB. PhD fellowship of the Research Foundation, Flanders.

Teaching Experience at University level

- At KU Leuven (Faculty of Engineering Science): Lecturer for the courses Distributed Systems; Comparative study of Programming Languages; Distributed Software Architectures: advanced study (Master Computer Science); Security & Privacy of Contemporary Distributed Systems (Advanced Master Cybersecurity);
- At Vrije Universiteit Brussel (Faculty of Sciences): Lecturer for the courses Distributed Systems (BS Computer Science, 2012-2014); Multicore Programming (MA Computer Science, 2011-2014); Capita Selecta of Software Engineering (MA Computer Science, 2009, 2012-2014).

Projects and Fund Raising

- 2023-present: Track Leader, Security Services & Platforms, Cybersecurity Research Program Flanders (large regional inter-university research programme, 8Meur/year).
- 2025-2029: FWO SBO Project “Scalable Software Development and Infrastructure for Self-sovereign Applications”. 1.86Meur.
- 2022: KU Leuven Start-up Grant. EUR 100K. ID: STG/22/022.
- 2017-2021: PI and project lead of a Nokia Bell Labs research project named “Future Software”. Project is internally funded (for an undisclosed amount) but has been running for 4+ years with 6-9 FTE staff.
- 2015-2019: Internal university funding: *Foundations of programming models for next-generation computing platforms*. Joint VUB SRP OZR-BOF grant with Prof. Dr. W. De Meuter. EUR 500K.
- 2013–2015: *ExaScience Life Lab*, project in partnership with Intel, Johnson&Johnson, imec and the 5 Flemish Universities (VUB, KUL, UA, UGent, UHasselt), partly funded by IWT. Budget: EUR 696K.
- 2013–2016: FWO project *Gradual verification of event-driven programs*. ID: G005813N (4-year research project as promotor, in collaboration with Prof. Dr. Bart Jacobs, KU Leuven). EUR 250K.
- 2013–2016: IWT grant *Een uniform programmeermodel voor rijke internettoepassingen*. ID: 121483 (as Scientific Promotor for 4-year PhD grant for Laure Philips).
- 2013: as main organizer of the “Vlaamse Programmeerwedstrijd”, raised EUR 18K sponsorship budget from industrial and academic partners to organize the 2013 edition of the contest at VUB.
- 2012–2015: IWT grant *Ontwikkeling van een bruikbaar en schaalbaar parallel programmeermodel voor many-core architectureen*. ID: 111436 (as Scientific Promotor for 4-year PhD grant for Mattias De Wael).
- 2011–2014: Post-doctoral Fellowship of the Research Foundation, Flanders (FWO) Renewal, *A unified computational model for ubiquitous cloud computing* ID: 1221312N (personal 3-year Post-doc grant).
- 2010–2012: joint IWT-Intel project *Intel ExaScience Lab*. ID: 090649 (as co-promotor). EUR 637K.

- 2009: FWO Travel Grant for 6-month stay at Google Research, Mountain View, California, USA.
- 2008–2011: Post-doctoral Fellowship of the Research Foundation, Flanders (FWO) *Abstraction and composition techniques for event-driven systems in Ubiquitous Computing*. ID: 1221309N (personal 3-year Post-doc grant).
- 2004–2008: PhD Fellowship of the Research Foundation, Flanders (FWO) *A Programming paradigm for Ambient Intelligence*. ID: 1188307N01 (personal 4-year PhD grant).

Scientific Promotorship and Mentoring Experience

Supervisor of PhD Students:

At KU Leuven:

- Start Oct 2025: Mr. Olav Blaak, “Secure Sandboxing for WebAssembly-based Smart Contracts”
- Start Nov 2025: Mr. Weihe Gao, “Distributed Software Frameworks for Decentralized Applications”
- Start Nov 2024: Mr. Glenn De Witte, “Programming languages for privacy-preserving and verifiable computing”
- Start Feb 2023: Ms. Weihong Wang, “Secure software stacks for decentralized applications” (CS Department FLOF fellowship)

At VUB:

- 2013–2014: Paul Bloüet (Joint PhD with Prof. Mario Südholt, École des Mines de Nantes, FR)
- 2013–2014: Steven Adriaensen (FWO Grant, co-promotor with Prof. Dr. Ann Nowé), graduated 2018.
- 2013–2014: Janwillem Swalens (IWT Grant), graduated 2019.
- 2012–2014: Laure Philips (IWT Grant)
- 2011–2014: Mattias De Wael (IWT Grant), graduated 2016.
- 2010–2014: Joeri De Koster (IWT Grant, co-promotor with Prof. Dr. Theo D’Hondt), graduated 2015.

2011–2014: Scientific Promotor of 1 Bachelors’ Thesis (12 ECTS) and 7 Masters’ Theses (24 ECTS).

2008–2014: Jury Member in the thesis committee of 14 MSc students.

2005–2011: Advisor to 6 PhD students, 8 MSc students and 2 BA students (under the promotorship of Prof. Dr. Theo D’Hondt and Prof. Dr. Wolfgang De Meuter).

Membership in PhD Thesis Committees

- 2025: Roozbeh Sarenche (KU Leuven COSIC, supervisor Prof. Bart Preneel), secretary
- 2024: Mahdi Sedaghat (KU Leuven COSIC, supervisor Prof. Bart Preneel)
- 2024: Tim Van Hamme (KU Leuven DistriNet, Prof. W. Joosen & Dr. Davy Preuveneers), secretary
- 2023: Kristof Jannes (KU Leuven DistriNet, Prof. W. Joosen & Prof. B Lagasse)
- 2023: Majid Salehi, KU Leuven COSIC. Supervisor: Prof. D. Hughes (KU Leuven).
- 2019: Bob Reynders, KU Leuven. Supervisors: Prof. F. Piessens (KU Leuven), Prof. D. Devriese (VUB).
- 2013: Nicolas Cardozo, Université Catholique de Louvain, Supervisors: Prof. K. Mens (UCL), Prof. T. D’Hondt (VUB)
- 2013: Stefan Marr, VU Brussel, Supervisors: Prof. T. D’Hondt (VUB), Dr. M. Haupt (Oracle)
- 2012: Yves Vandriessche, VU Brussel, Supervisors: Prof. T. D’Hondt (VUB), Dr. E. D’Hondt (VUB)

Membership in University Councils

At KU Leuven:

- 2023-: member, POC (Education Committee) Master Computer Science
- 2023-: observing member, POC ((Education Committee) Bachelor Computer Science
- 2023-: observing member, POC ((Education Committee) Advanced Master Cybersecurity
- 2023-: member, Departmental Council, Computer Science
- 2023-: member, Faculty Council, Engineering Science

At VUB:

- 2010–2014: member, VUB Computer Science Department council
- 2010–2014: member, VUB Faculty of Sciences council
- 2012–2014: member, VUB Bachelor in CS curriculum council
- 2013: member, hiring committee for Professorship position in Web & Information Systems

Research Stays Abroad

- 2009–2010: **Google Research**, Mountain View, California, **USA** (6 months). Host: Dr. Mark S. Miller.
- 2009: Information Technology Center, **University of Tokyo, Japan** (2 weeks). Host: Prof. A. Yonezawa.
- 2008: ADAM Group, **INRIA**, Lille, **France** (2 weeks). Host: Prof. Stéphane Ducasse.

- 2007: PLEIAD Lab, **Universidad de Chile**, Santiago, **Chile** (1 month). Host: Prof. Éric Tanter.
- 2007: LAMP Group, **EPFL**, Lausanne, **Switzerland** (1 week). Host: Prof. Martin Odersky.

Membership in Academic, Educational and Industry Organizations

- 2011–present: Founding Member of the [IFIP Working Group 2.16](#) on Programming Language Design.
- 2025: Board member, secappdev.org VZW. secappdev.org is a non-profit organization that aims to broaden security awareness in the development community and advance secure software engineering practices. It runs a yearly weeklong SecAppDev course.
- 2018–2019: represented Nokia Belgium at Flemish AI policy and Flanders AI Research Program meetings organised by Agoria, VLAIO, imec and the Ministry of Economy, Science & Innovation.
- 2016–2019: Industrial Advisory Board member, VLAIO Tearless SBO Project of Prof. Frank Piessens (DistriNet KULeuven), Prof. Elisa Gonzalez Boix (SOFT-VUB), Prof. Coen De Roover (SOFT-VUB).
- 2010–2015: Member of [ECMA International Technical Committee 39](#), the standardisation committee of the ECMAScript (JavaScript) programming language.
- 2010–2014: Member of the scientific steering committee of the [Intel ExaScience Europe Lab](#).
- 2011–2014: Main organizer (2013), Chairman (2012) and Jury Member (2014, 2012, 2011) of the [Flemish Programming Contest](#) (Vlaamse Programmeerwedstrijd).
- 2010–2014: Member of the organizing committee of the [Belgian Olympiad in Informatics](#) (beOI VZW).

Organizer of International Workshops and Symposia

- 2024: co-organiser, Industry Seminar on “The Future of Blockchain Platforms”, May 31st, 2024 (hosted at KU Leuven) (51 participants)
- 2023: co-organizer and speaker: seminar on Blockchain & Distributed Ledgers: Challenges for Secure Data Management, August 29 2023, Leuven.
- 2023: Main organizer: Joint DistriNet-COSIC-Nokia Bell Labs [Workshop on Security, Privacy & Verifiable Computing](#), June 15 2023, Leuven (52 registered participants).
- 2018: Workshop on [Programming Technology for the Web](#) (ProWeb), Brussels, Belgium. Co-organizer.
- 2018: Hosted 9th IFIP [Working Group on Programming Language Design meeting](#) in Antwerp, Belgium.
- 2014: Workshop on [Dynamic Languages and Applications](#) (Dyla), Edinburgh, UK. Co-organizer.
- 2013: Workshop on [Secure Cloud and Reactive Internet Programming Technology](#), Brussels.
- 2012: First [Erlang Factory Lite Brussels](#) Symposium, Brussels, Belgium.
- 2010: Workshop on [Decentralized Coordination of Distributed Processes](#), Amsterdam, Netherlands.
- 2009: Workshop on [Distributed Objects for the 21st Century](#), ECOOP '09, Genova, Italy.

Program Committee Member (international conferences and symposia)

- 2025: <Programming> 2026 Conference
- 2025: Reviewer for the Journal “The Art, Science, and Engineering of Programming”
- 2025: IEEE ICBC 2026 Conference (flagship IEEE conference on Blockchain and Cryptocurrencies)
- 2024: IEEE ICBC 2025 Conference (flagship IEEE conference on Blockchain and Cryptocurrencies)
- 2024, 2023: Annual Privacy Forum (APF)
- 2022: ACM Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software ([Onward! Papers](#))
- 2021: IEEE Secure Development Conference ([SecDev](#))
- 2018: WWW (The Web Conference), [Web Programming](#), Design, Analysis and Implementation track
- 2018: IEEE SCAM, Int’l Working Conf. on Source Code Analysis and Manipulation ([Engineering track](#))
- 2017: OOPSLA 2017, Int’l Conf. on Object-oriented Programming, Systems, Languages & Applications
- 2015: ECOOP 2015, the European Conference on Object-oriented Programming.
- 2014: ACM Conference on New Ideas in Programming and Reflections on Software ([Onward!](#)).
- 2012: ACM SIGPLAN Dynamic Languages Symposium ([DLS](#)).
- 2012: ACM Conference on New Ideas in Programming and Reflections on Software ([Onward!](#)).
- 2011: ACM Conference on New Ideas in Programming and Reflections on Software ([Onward!](#)).

Program Committee Member (international workshops)

- 2024, 2023: ACM EdgeSys Workshop at EuroSys
- 2023, 2022, 2020: Workshop on Reactive and Event-based Languages & Systems ([REBLS](#)), co-located with ACM SPLASH.
- 2020: Workshop on ENgineering Intelligent Applications' Code ([ENIAC](#)), ACM <Programming> 2020.
- 2020, 2019, 2018, 2017, 2016: ACM SIGPLAN Workshop on Meta-Programming Techniques and Reflection, co-located with ACM SPLASH ([META 2019](#)).
- 2019, 2017: Workshop on Programming Technology for the Future Web ([ProWeb 2019](#)).
- 2016, 2015, 2014, 2013, 2012: ACM SIGPLAN Workshop on Agents and Actors Reloaded ([AGERE!](#)).
- 2015: Workshop on the Future of Programming, co-located with OOPSLA 2015 ([FPW 2015](#)).

- 2014: Scala 2014 Workshop, co-located with ECOOP 2014 ([Scala 2014](#)).
- 2012: Workshop on Relaxing Synchronization for Multicore and Manycore Scalability ([RACES](#)).
- 2011: Workshop on Dynamic Languages and Applications ([DYLA](#)).
- 2011: Symposium on Ubiquitous Computing and Ambient Intelligence ([UCAmi](#)).
- 2010: 4th Argentinian Smalltalk Conference ([Smalltalks](#)).

External Reviewer for INRIA Evaluation Committee (2021) • ECOOP (2018, 2012, 2011, 2007, 2006, 2005) • Modularity (2014) • ESOP (2013) • OOPSLA (2013) • Elsevier Computer Languages, Systems & Structures (2011) • Elsevier Science of Computer Programming (2013, 2012, 2011, 2009) • IEEE Software (2010) • IEEE Trans. Softw. Eng. (2008) • ACM DLS (2009) • ACM SAC (2006) • MODELS (2006) • Edit. Board Member, Intl. Journal of Handheld Computing Research (2009-2013)

Guest Lectures at other universities

- 2019: *Machine learning at Nokia*, “Big Data Science” MA-level course at **Ghent University**, Prof. Tijl De Bie.
- 2017: *Introduction to Distributed Systems*, “Software Engineering” BA-level course at **Ghent University**, Prof. Christophe Scholliers.
- 2016: *Microservices and Docker: from theory to practice*, postgraduate seminar on microservices organised by **KULeuven KULAK** Postgraduate Centre.
- 2016: *The JavaScript Meta-object Protocol*, “Programming Paradigms” MA-level course at **UC Louvain**, Prof. Kim Mens.

Invited Talks, Keynote Talks, Industry meeting Talks

A full list including conference talks at academic conferences is available at <https://tvcutsem.github.io/talks/>

- 2024, Invited lecture, “A gentle introduction to Ethereum and smart contracts”, SecAppDev 2024, June 3-7, Leuven.
- 2021, Accepted Talk at Developer Conference: *Architecting robust JS apps: a practitioner’s guide to Secure ECMAScript*, **Techorama**.
- 2019, Invited Talk: *Architecting robust JavaScript applications*, **Peopleware** ICTech Day (company seminar), Antwerp, Belgium.
- 2018, Invited Talk: *Will software developers soon be replaced by AI? The sense and nonsense of Artificial Programming*, **FWO Kennismakers** (science policy conference), Antwerp, Belgium.
- 2017, Keynote: *From Documents to Dialogues: Programming Technology for the Conversational Web*. Workshop on Programming Technology for the Future Web (**ProWeb 2017**), Brussels, Belgium.
- 2016, Keynote: *Control Flow Goodness in ECMAScript 2015 and Beyond*, [jsconf.be](#) - the Belgian JavaScript developer conference, Bruges, Belgium.
- 2016, Invited Talk: *Writing Robust JavaScript Code using Modern JavaScript*, Secure Application Development (**SecAppDev**) course seminar and Belgium OWASP Chapter meetup, Leuven, Belgium.
- 2016, Invited Talk: *ECMAScript 2015: the Future of JavaScript is Now!*, **SPLASH** 2015 Industry Track, Pittsburgh, USA.
- 2015, Invited Talk: *ECMAScript 6: the Future of JavaScript is Now!*, **Ordina** JOIN (company seminar), Mechelen, Belgium.
- 2015, Keynote: *The road to ES6, and beyond: a tale about JavaScript’s past, present and future*, [jsconf.be](#) - the Belgian JavaScript developer conference, Bruges, Belgium.
- 2014, Invited Talk: *ECMAScript 5 and 6: the present and future of JavaScript*, **Cegeka** (company seminar), Leuven, Belgium.
- 2014, Invited Talk: *Membranes as Ownership Boundaries*, **Dagstuhl** Seminar on “Scripting Languages and Frameworks: Analysis and Verification”, Dagstuhl, Germany.
- 2014 and 2013, Invited Talk: *Writing Robust JavaScript Code*. Secure Application Development (**SecAppDev**) seminar, Leuven, Belgium.
- 2012, Invited Talk: *The Javascript Meta-object Protocol*. Summer School of the 26th European Conference on Object-oriented Programming (**ECOOP**), July 2012, Peking, China.
- 2012, Keynote: *AmbientTalk: Modern actors for Modern Networks*. 14th Intl. Workshop on Formal Techniques for Java-like Programs (**FTfJP**), co-located with ECOOP and PLDI 2012, Peking, China.
- 2010, Invited Talk: *AmbientTalk: A scripting language for mobile phones*. “Emerging Languages Camp” session at the O’Reilly Open Source Conference (**OSCON**), 2010, Portland, USA.
- 2009, Invited Talk: *Ambient-oriented Programming and AmbientTalk*. **Japan National Institute of Informatics**, 25th GRACE Seminar on Advanced Software Engineering, Tokyo, Japan.
- 2009, Invited Talk, *Ambient-oriented Programming*. Alcatel-Lucent **Bell Labs** (company seminar), Antwerp, Belgium.
- 2006, Keynote: *Ambient-oriented Programming*. First workshop on Software Engineering for Pervasive

Services (SEPS), in conjunction with IEEE ICPS 2006, Lyon, France.

Prizes and Awards

- 2025: **Best lecturer** in Master Computer Science, award by the Engineering students Student Circle.
- 2024: **Best Paper** award at BRAINS 2024 Conference.
- 2019: **Best Presentation** Award, Nokia AI Conference.
- 2018: **Best Poster** Award, Nokia AI Conference.
- 2017: **Most Notable Paper Award**, Dynamic Languages Symposium for our 2007 paper “Mirages: Behavioral Intercession in a Mirror-based Architecture”.
- 2017: **Best Demo Award**, ACM conference on distributed and event-based systems (DEBS) for our demonstrator “Building Connected Car Applications on top of the World-Wide Streams Platform”.
- 2008: **Best Paper Award**, International Workshop on Middleware for Pervasive and ad-hoc Computing at Middleware 2008.
- 2004: recipient of the VUB Faculty of Science’s **Science Award** (Dutch: “Prijs van de Wetenschappen”), awarded to the graduated student with the overall best grades over their entire course of study.

Media mentions of my work in popular press and dissemination towards software practitioners

- 2019: in celebration of the [50th anniversary](#) of UNIX at Bell Labs, I designed [The Unix Game](#) which got mentioned on [Hacker News](#) and had global impact (15.000+ registered players from 144 countries).
- 2011: Front-page story covering my essay “[Why programming languages?](#)” on “Lambda the Ultimate”, the Programming Languages Weblog, June 27, 2011.
- 2011: Article on PhD work: “[Scripttaal voor mobiele apps](#)”, Datanews No. 3, February 2011 (in Dutch).
- 2010: [Google Tech Talk](#) on my JavaScript language design work at Google (7900+ views).
- 2010: My JavaScript library traits.js featured on the website [Ajaxian.com](#) (April 13th, 2010).
- 2010: Article in MIT Technology Review Magazine, July 2010, “[New Languages, and Why We Need Them](#).” Also appeared in Dutch in Datanews, July 2010, “[Nood aan nieuwe talen én aan meertaligheid](#)”.
- 2010: [interview on Channel 9](#) (40.000+ views) after an invited talk at the Emerging Languages Camp (see “Invited Talks”). Channel 9 is a Microsoft video channel for software developers.

Involvement in Open Source software development

- 2020: project lead, neural code search engine ([GitHub](#)).
- 2019: project lead, Code Compass, a search engine for software packages ([GitHub](#), [Website](#)).
- 2012-2018: author, reflect.js, a JavaScript library for reflection ([GitHub](#)).
- 2010-2015: author, traits.js, a JavaScript library for object composition using traits ([Website](#)).
- 2006-2014: author, Java implementation of the AmbientTalk programming language ([GitLab](#), [Website](#)).
- 2011-2014: author, stm-in-clojure, a Clojure library for software-transactional memory ([GitHub](#)).
- 2011-2012: author, clj-worlds, a Clojure library for scoped side-effects ([GitHub](#)).
- 2010-2012: author, an ECMAScript 5 compliant parser written in JavaScript ([GitHub](#)).

Patents granted

- [EP3468150B1](#). “Handling questions of conversational services”, filed with EPO 04-10-2017.

Patents filed

- [GB2601310A](#). “Methods and apparatuses relating to object identification”, filed UK PO 01-06-2022.
- [EP4109299A1](#). “Interactive generation of queries for timed annotated media items”, EPO 24-06-2021.
- [EP3474510A1](#). “Granting to a device access to an access point”, filed with EPO 20-10-2017.
- [EP3474159A1](#). “Method and system to determine capabilities of bots”, filed with EPO 17-10-2017.

Publications

H-index [according to Google Scholar](#): 24

Online publication lists: [Google Scholar](#) • [DBLP](#) • [ACM Digital Library](#)

Top 5 representative papers are marked with [★]. Papers with 50+ citations are marked with [50+], papers with 100+ citations are marked with [100+] (citation count source: Google Scholar as of August 2021).

Selected articles in peer-reviewed international journals

1. Xu Y, Slaats T, Döder B, Troels Hildebrandt T, **Van Cutsem T**. Safe design and evolution of smart contracts using dynamic condition response graphs to model generic role-based behaviors. *Journal of Software Evolution and Process*. 2025; 37(1):e2730. <https://doi.org/10.1002/smr.2730>
2. J. De Koster, S. Marr, **T. Van Cutsem**, Theo D'Hondt. [Domains: Sharing State in the Communicating Event-Loop Actor Model](#). *Computer Languages, Systems & Structures*, Vol. 45, pp. 132-160, April 2016, Elsevier.
3. [50+] M. De Wael, S. Marr, B. De Fraine, **T. Van Cutsem**, W. De Meuter. [Partitioned Global Address Space Languages](#). *ACM Computing Surveys*, Vol 47, no. 4, article 62, July 2015, 27 p.
4. **T. Van Cutsem**, M. S. Miller. [Robust Trait Composition for Javascript](#). *Science of Computer Programming*, Vol. 98, Part 3, February 2015, pp. 422-438, Elsevier.
5. J. De Koster, S. Marr, T. D'Hondt, **T. Van Cutsem**. [Domains: Safe sharing among actors](#). *Science of Computer Programming*, Vol. 98, Part 2, February 2015, pp. 140-158, Elsevier.
6. **T. Van Cutsem**, E. Gonzalez Boix, C. Scholliers, A. L. Carreton, D. Harnie, K. Pinte, W. De Meuter. [AmbientTalk: programming responsive mobile peer-to-peer applications with actors](#). *Computer Languages, Systems & Structures*, Vol. 40, issues 3-4, December 2014, pp. 112-136, Elsevier.
7. [★] [100+] E. Bainomugisha, A. Lombide Carreton, **T. Van Cutsem**, S. Mostinckx, W. De Meuter. [A Survey on Reactive Programming](#). 39 p. *ACM Computing Surveys*, Vol. 45, no. 4, Article 52, August 2013, 34p. pages. SCI 5-year impact factor: 9.169.
8. [★] [100+] **T. Van Cutsem**, M. S. Miller. [Proxies: Design Principles for Robust Object-oriented Intercession APIs](#). *ACM SIGPLAN Notices* Vol. 45 No. 12, pp. 59-72, December 2010. Also published in Proceedings of the ACM Dynamic Languages Symposium, pp. 59-72, ACM, NY. Acceptance rate: 28%.
9. [★] S. Mostinckx, **T. Van Cutsem**, S. Timbermont, E. Gonzalez Boix, E. Tanter, W. De Meuter. [Mirror-based Reflection in AmbientTalk](#). *Software: Practice & Experience*, Vol. 39 No. 7, pp. 661-699, May 2009. Wiley.
10. **T. Van Cutsem**, S. Mostinckx, W. De Meuter. [Linguistic Symbiosis between Event Loop Actors and Threads](#). *Computer Languages, Systems & Structures*. Vol. 35 No. 1, pp. 80-98. April 2009, Elsevier.
11. J Vallejos, **T. Van Cutsem**, E. Gonzalez Boix, S. Mostinckx, W. De Meuter: [The Message-Oriented Mobility Model](#). *Journal of Object Technology*, vol. 6, no. 9, pp. 363-382, October 2007.

Editor of proceedings

1. **T. Van Cutsem**, M. S. Miller (Eds.) [Proceedings of the First international Workshop on Decentralized Coordination of Distributed Processes](#) (Amsterdam, The Netherlands, June 2010) Electronic Proceedings in Theoretical Computer Science, Vol. 27, 36 p. ISSN: 2075-2180.
2. **T. Van Cutsem**, J. Fox, O. Lehrmann Madsen, E. Jul, G. Bracha (Eds.) [Proceedings of the First International Workshop on Distributed Objects for the 21st Century](#) (Genova, Italy, July 2009) ACM International Conference Proceedings Series, ACM Press, 26 p., ISBN: 978-1-60558-545-1.

Selected articles in peer-reviewed proceedings of conferences and symposia

1. W. Wang and **T. Van Cutsem**, "Depermissioning Web3: a Permissionless Accountable RPC Protocol for Blockchain Networks," 2025 IEEE 45th *International Conference on Distributed Computing Systems* (ICDCS), Glasgow, United Kingdom, 2025, pp. 210-220. CORE Rank-A.
2. R. Hias, W. Wang, J. Vanhoof and **T. Van Cutsem**, "A Scalable State Sharing Protocol for Low-Resource Validator Nodes in Blockchain Networks," 2024 6th *Conference on Blockchain Research & Applications for Innovative Networks and Services* (BRAINS), Berlin, Germany, 2024, pp. 1-9. Best Paper.
3. J. Surmont, W. Wang, **T. Van Cutsem**. Static Application Security Testing of Consensus-Critical Code in the Cosmos Network. In *Proceedings of the 5th Conference on Blockchain Research & Applications for Innovative Networks and Services* (BRAINS '23), November 2023, IEEE. To appear.
4. G. Heyman, R. Huysegems, P. Justen, **T. Van Cutsem**. [Natural Language-guided Programming](#). *Proc. ACM SIGPLAN Symposium on New Ideas in Programming and Reflections on Software (Onward!)*, 2021. Association for Computing Machinery, New York, NY, USA, pp. 39-55, 2021.
5. B. Theeten, F. Vandeputte, **T. Van Cutsem**. [Import2vec: learning embeddings for software libraries](#). *Proc. 16th International Conference on Mining Software Repositories (MSR '19)*, pp. 18-28, 2019, IEEE Press.
6. L. Philips, C. De Roover, **T. Van Cutsem**, W. De Meuter. [Towards Tierless Web Development without](#)

- [Tierless Languages](#). Proc. of the 2014 ACM International Symposium on New Ideas, New Paradigms, and Reflections on Programming & Software (Onward! 2014). ACM, pp. 69–81, 2014.
7. M. De Wael, S. Marr, **T. Van Cutsem**. [Fork/join parallelism in the wild: documenting patterns and anti-patterns in Java programs using the fork/join framework](#). *Proc. of the 2014 International Conference on Principles and Practices of Programming on the Java platform (PPPJ '14)*. ACM, pp. 39–50, 2014.
 8. [★] [50+] **T. Van Cutsem**, M. S. Miller. [Trustworthy Proxies: Virtualizing Objects with Invariants](#). *Proc. European Conference on Object-oriented Programming (ECOOP 2013)*, LNCS Vol. 7920, pp. 154–178, Springer, 2013. Acceptance rate: 24%.
 9. M. S. Miller, **T. Van Cutsem**, B. Tulloh. [Distributed Electronic Rights in JavaScript](#). Invited paper, *Proc. European Symposium on Programming (ESOP 2013)*, pp. 1–20, LNCS Vol. 7792, Springer, 2013.
 10. M. De Wael, D. Ungar, **T. Van Cutsem**. [When spatial and temporal locality collide: the case of the missing cache hits](#). In *Proceedings of the 4th ACM/SPEC International Conference on Performance Engineering (ICPE '13)*, pp. 63–70, ACM, 2013. Acceptance rate: 37%.
 11. E. Gonzalez Boix, C. Noguera, **T. Van Cutsem**, W. De Meuter, T. D'Hondt. [REME-D: a Reflective, Epidemic Message-Oriented Debugger for Ambient-Oriented Applications](#). *Proc. 26th ACM Symposium on Applied Computing (SAC 2011)*, pp. 1275–1281, ACM, March 2011. Acceptance rate: 37%.
 12. E. Gonzalez Boix, A. Lombide Carreton, C. Scholliers, **T. Van Cutsem**, W. De Meuter, T. D'Hondt. [Flocks: Enabling Dynamic Group Interactions in Mobile Social Networking Applications](#). *Proc. 26th ACM Symposium on Applied Computing (SAC 2011)*, pp. 425–432, ACM, 2011. Acceptance rate: 25%.
 13. A. Lombide Carreton, S. Mostinckx, **T. Van Cutsem**, W. De Meuter. [Loosely-coupled Distributed Reactive Programming in Mobile Ad Hoc Networks](#). In *proceedings of the 48th International Conference on Objects, Models, Components, Patterns (TOOLS EUROPE 2010)* (Malaga, Spain, June 2010), Jan Vitek (Ed.), LNCS, pp. 41–60, Springer-Verlag. Acceptance rate: 28%.
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2. **T. Van Cutsem** and P. Haller. Distributed and Parallel Computing. In [Actors in Scala: Concurrent programming for the multi-core era](#), P. Haller and F. Sommers, pp. 123-140, Artima Press, January 2012, 187 p. *Chapter contributed by invitation, not peer-reviewed.*
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4. **T. Van Cutsem**, W. De Meuter. [Event-driven Mobile Computing with Objects](#). In *Principles and Applications of Distributed Event-Based Systems*, pp. 324-345, A. Hinze and A. Buchmann (Eds.) IGI Global, 2010.
5. S. Mostinckx, J. Dedecker, E. Gonzalez Boix, **T. Van Cutsem**, W. De Meuter. [Ambient-oriented Exception Handling](#). *Advances in Exception Handling*, LNCS Vol. 4119, C. Dony, J.L. Knudsen, A. Romanovsky, A. Tripathi (Eds.), pp. 141-160, 2006, Springer.

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2. A. Jeffrey, **T. Van Cutsem**. [Functional Reactive Programming with nothing but Promises](#). In *Workshop on Reactive and Event-based Languages & Systems (REBS)*. Pittsburgh, US, 2015.
3. T. Coppieters, L. Philips, W. De Meuter, **T. Van Cutsem**. [An open implementation of cloud types for the web](#). *Proc. First Workshop on Principles and Practice of Eventual Consistency (PaPEC '14)*, co-located with EuroSys'14, pp. 1-2, ACM, 2014.
4. J. Swalens, S. Marr, J. De Koster, **T. Van Cutsem**. [Towards Composable Concurrency Abstractions](#). In *Workshop on Programming Language Approaches to Concurrency and Communication-cEntric Software (PLACES 2014)*, co-located with ETAPS, EPTCS, pp. 54-60, January 2014.
5. J. De Koster, S. Marr, T. D'Hondt, **T. Van Cutsem**. [Tanks: multiple reader, single writer actors](#). In *Proc. 2013 workshop on Programming based on actors, agents, and decentralized control (AGERE! 2013)*, co-located with ACM SPLASH 2013, pp. 61-68, ACM, 2013.
6. J. De Koster, **T. Van Cutsem**, T. D'Hondt. [Domains: safe sharing among actors](#). *Proc. 2nd edition on Programming systems, languages and applications based on actors, agents, and decentralized control abstractions (AGERE! 2012)*, co-located with ACM SPLASH 2012, pp. 11-22, ACM, 2012.
7. S. Marr, J. Nicolay, **T. Van Cutsem**, T. D'Hondt. [Modularity and conventions for maintainable concurrent language implementations: a review of our experiences and practices](#). *Proc. of the 2012 workshop on Modularity in Systems Software (MISS '12)*, pp. 21-26, ACM, 2012.
8. M. S. Miller, **T. Van Cutsem**. [Languages for Debuggable Distributed Algorithms](#). *Presented at the 2012 Languages for Distributed Algorithms (LADA) workshop*, co-located with ACM POPL 2012.
9. **T. Van Cutsem**, M. S. Miller. [Traits.js: robust object composition and high-integrity objects for ecmascript 5](#). *Proc. of the 1st ACM SIGPLAN international workshop on Programming language and systems technologies for internet clients (PLASTIC '11)*, pp. 1-8, ACM, 2011.
10. T. Suzuki, K. Pinte, **T. Van Cutsem**, W. De Meuter, A. Yonezawa. [Programming Language Support for Routing in Pervasive Networks](#). *Proc. 8th International IEEE Workshop on Middleware and System Support for Pervasive Computing (PerWare 2011)*, IEEE, March 2011.
11. **T. Van Cutsem**, S. Marr, W. De Meuter. [A language-oriented approach to Teaching Concurrency](#). *International Workshop on Curricula for Concurrency and Parallelism*, co-located with SPLASH 2010.
12. P. Costanza, B. De Fraine, **T. Van Cutsem**. [Improving the Data Locality of Work Stealing: a Domain-specific Approach](#). *International Workshop on Concurrency for the Application Programmer*, co-located with ACM SPLASH 2010.

13. A. L. Carreton, D. Harnie, E. G. Boix, C. Scholliers, **T. Van Cutsem**, W. De Meuter. [Urbiflock: An experiment in dynamic group management in Pervasive social applications](#). *8th IEEE International Conference on Pervasive Computing and Communications Workshops (PERCOM)*, pp. 250-255, IEEE, 2010.
14. H. Schippers, **T. Van Cutsem**, S. Marr, M. Haupt, R. Hirschfeld. [Towards an actor-based concurrent machine model](#). *Proc. of the 4th workshop on the Implementation, Compilation, Optimization of Object-Oriented Languages and Programming Systems (ICOOOLPS '09)*, co-located with ECOOP 2009, pp. 4-9, ACM, 2009.
15. A. Lombide, **T. Van Cutsem**, W. De Meuter. [Reactive queries in mobile ad hoc networks](#). *Proc. of the 6th international workshop on Middleware for pervasive and ad-hoc computing (MPAC '08)*, co-located with ACM Middleware 2008, pp. 13-18, 2008. Best Paper Award. Acceptance Rate: 32%.
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18. J. Dedecker, **T. Van Cutsem**, S. Mostinckx, W. De Meuter, T. D'Hondt. [AmbientTalk: Language Support for Mobile Computing](#). *Proc. Workshop on System Support for Future Mobile Computing Applications (FUMCA 06)* (Orange County, USA, September 2006), pp. 35-42, IEEE, 2006.
19. D. Standaert, E. Tanter, **T. Van Cutsem**. [Design of a Multi-Level Reflective Architecture for Ambient Actors](#). *Proc. 2nd workshop on Object Technology for Ambient Intelligence and Pervasive Computing: Language Constructs and Infrastructures (OT4AmI)* co-located with ECOOP '06, July 2006.
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1. W. Van Raemdonck, **T. Van Cutsem**, K. Sheykh Esmaili, M. Cortes, P. Dobbelaere, L. Hoste, E. Philips, M. Roelands, L. Trappeniers. [Building Connected Car Applications on Top of the World-Wide Streams Platform](#). *Proceedings of the 11th ACM International Conference on Distributed and Event-based Systems (DEBS '17)*. pp. 315-318, 2017, ACM. Best Demo Award.
2. **T. Van Cutsem**, [AmbientTalk: Modern Actors for Modern Networks](#). *Presented at the AGERE!'11 workshop. In SPLASH 2011 Workshop Proceedings*, pp. 227-230, ACM, October 2011.
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